Measurements of triggering stress transmitted through the upper snow cover

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ABSTRACT: We present data describing how stress from human triggers penetrates the snow cover. In the majority of fatal avalanches, skiers and snowmobilers apply load to the snow cover that triggers the initial failure in a weak layer. Understanding how stresses from dynamic surface loads transmit through the snow cover can help people avoid situations where they can trigger avalanches. Capacitive sensors were used to measure this dynamic stress within the mountain snow cover. The sensors were used on 33 separate field days to collect over 1,420 measurements of loading by skiers and snowmobiles. We described the snow cover for the experiments according to a *bridging index.* This allowed us to quantitatively analyze how the stress added by a skier or snowmobile transmits though the snow cover. This added stress was then related to localized slope stability using stability indices. Static stress calculations for an elastic homogeneous 2D snow cover were calibrated to show measured stress values of loading caused by the passage of a snowmobile. The change in shape and magnitude of the induced stress was plotted for three "typical" snow cover structures. Relatively soft snow allowed the specific levels of dynamic stress to penetrate deeper into the snow cover, thus increasing the probability of initiating a failure in a weak layer. Whereas, supportive snow limited the depth that the dynamic stress penetrated by spreading it out laterally.

Keywords: Stress, localized dynamic loading, snow properties, snow stability evaluation, bridging strength, bridging index, snowmobile.

1. INTRODUCTION

Most snow avalanche fatalities result from people triggering the avalanches themselves (Schweizer and Lütschg, 2001; Jamieson et al., 2010; Harvey et al., 2012). Backcountry skiing, snowmobiling, snowboarding, etc. applies localized dynamic loads (LDL) to the snow cover, which can initiate failures in weak layers and possibly trigger avalanches. This loading imparts stress to the snow cover and depends mainly on: a) the type of trigger creating the load and b) the medium that the resulting stress travels through. Understanding how the stress from different loading types transmits through the mountain snow cover may help people avoid situations where they can trigger avalanches.

Overall, we build on Föhn's (1987) stability indices that were modified for ski penetration by Jamieson and Johnston (1998). We utilize concepts from the finite element work done by Schweizer (1993), Jones et al. (2006) and Habermann et al. (2008). And, we compare results with the load cell measurements by Schweizer et al. (1993, 2001). Finally, we use the concept of *bridging index* first proposed by Schweizer and Jamieson (2003).

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Figure 1: Skier, Mike Wheater, loading the snow surface above the buried sensors.

We are preparing a peer-reviewed paper that will more thoroughly discuss the important work preceding this study.

In this paper the bridging index is used to quantify the effect of snow cover stratigraphy on the transmission of stress through the upper snow cover. We then relate this additional stress to localized slope stability using an empirical model for shear strength of buried surface hoar (Zeidler and Jamieson, 2006).

2. METHODS

We used capacitive sensors mounted to 1.2 m long aluminum sheets inserted into the snow cover to measure the additional stress from human-induced loading. Refer to Thumlert et al. (2012, 2013) for more detail on the methods used to collect stress data.

The data presented in this paper were from loading performed on the surface of the snow by either:

- skier sliding straight downhill and bending their knees over the sensors to simulate a typical ski turn
- snowmobile driving uphill over the sensors

Figure 1 shows a skier loading the snow surface above the buried sensors.

In order to quantify the snow cover for analysis, we classified it according to a *bridging index,* similar to Schweizer and Jamieson (2003). The *index* was defined as the hardness of a particular layer in the snow cover (Geldsetzer and Jamieson, 2001) multiplied by the thickness of that layer. The *bridging index* was calculated for the layers above each sensor and summed. Thus, a sensor inserted at 80 cm into the snow cover would be assigned the total bridging index for all the layers between 0 cm and 80 cm.

To relate the measured dynamic stress to slope stability we assume the initial failure in a weak layer is a shear failure (Reiweger and Schweizer, 2010). We then converted the measured normal stress to shear stress ($\Delta\tau_{zx}$) using a calculated ratio based on slope angle from each day's measurements. The ratio was obtained from calculations on a homogenous elastic isotropic material (Das, 1983). Details of this calculation will be available in a future paper or by contacting the authors.

Jamieson (1995) showed that most skier-triggered dry slab avalanches occur when the skier stability index adjusted for skier penetration (S_{k38}) is less than 1.5. Thus, we use the skier stability index as a proxy for skier triggering on the slope scale.
For each dynamic stress measurement, we calculated Föhn's (1987) skier stability index:

$$S_k = \frac{\tau_s}{\tau_{zx}+\Delta\tau_{zx}} \qquad (1)$$

where:

$$\tau_s = 1.51 * Load^{0.93} \qquad (2)$$

where τ_s represents the shear strength of a hypothetical layer of buried surface hoar obtained by regression (*$R^2 = 0.82$, $p < 1.0^{-17}$*) from Zeidler and Jamieson (2006). *Load* was defined as the weight per unit area of the overlying snow (kPa).

$$\tau_{zx} = \rho g h \sin\varphi \cos\varphi \qquad (3)$$

where τ_{zx} represents the shear stress in the weak layer due to the overlying slab, ρ is density of slab, *g* is gravity, φ is the slope angle and *h* is thickness of slab measured vertically.

$\Delta\tau_{zx}$ represents the added shear stress from skiing or snowmobiling.

3. DATA

In total, 36 days were used for data collection in over four winter seasons. In this study we analyzed 1,420 measured localized loading events by skiing or snowmobiling. We measured and analyzed vertical depth of sensor below undisturbed snow surface, type of localized dynamic load (e.g. skier, snowmobile), penetration depth of trigger into the snow cover, effective depth (defined as penetration depth subtracted from depth of sensor), snow cover layer density and snow cover hand hardness.

4. RESULTS

The measurements shown in Figures 2 through 5 display shear stress added to the snow cover by skiers and snowmobiles according to the *bridging index.* Figures 2 (skier data) and 3 (snowmobile data) show only the additional shear stress added by skiers or snowmobiles, whereas Figures 4 and 5 add the shear stress due to the overlying slab, the shear strength of surface hoar and the corresponding stability indices.

The data in Figures 2 and 3 were fit with a power law using the non-linear least squares method (Bates and Chambers, 1992) according to the following equation:

$$\tau = (BI * a)^b \qquad (4)$$

with τ = stress, *BI* = bridging index and the fitted values a and b. The fitted values for the skier data (Figure 2) were a =-7.7 x 10^{-3} and b = -1.5 and yielded **R^2 = 0.26** for 454 skier data. The constants for the snowmobile data (Figure

3) were a = -5.6×10^{-3} and b = -1.5 and yielded **R^2 = 0.40** for 515 snowmobile data. Note, when the shear stress values were plotted against depth and fitted with the same power law least squares regression the **R^2** values were **0.16** for skier data and **0.31** for snowmobile data and (results not shown).

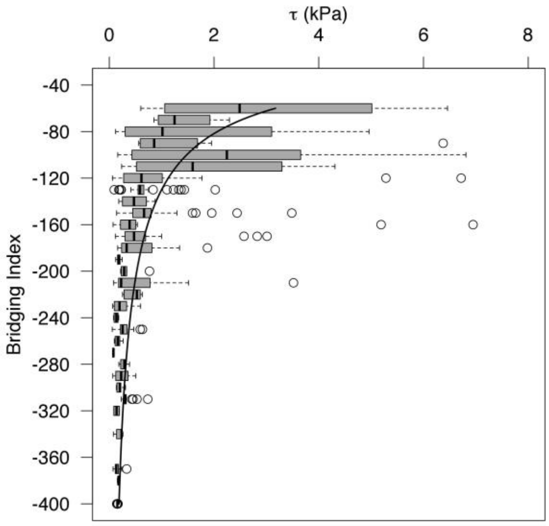

Figure 2: Box plots of shear stress by bridging index based on measurements of normal stress for skiers. The curved line indicates the fitted model. Boxes span the interquartile range. Whiskers extend to 1.5 times the interquartile range. Open circles indicate outliers.

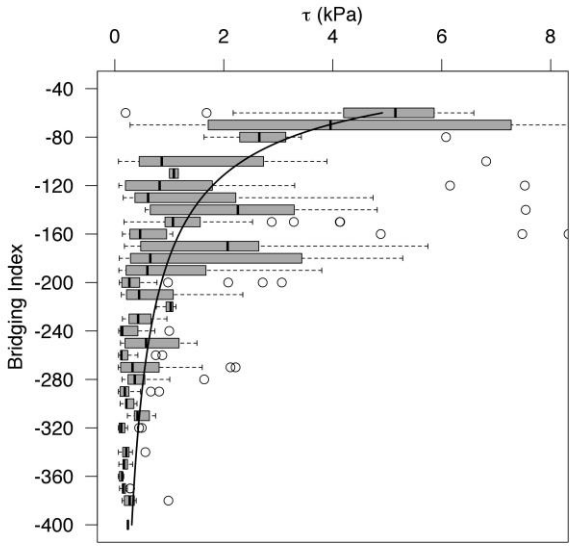

Figure 3: Box plots of shear stress by bridging index for measurements of the passage of snowmobiles. The curved line indicates the fitted model. Boxes, whiskers and open circles as in Figure 2.

Figures 4 and 5 show the main variables used to calculate the skier stability index plotted against the bridging index. Shear stress from skiing or snowmobiling is added to the shear stress due to the overlying snow and combined with the hypothetical shear strength of buried surface hoar to calculate skier stability index values. The linear regression for skier data between bridging index and S_k produced an $R^2 = 0.53$ and $p < 2.2 \times 10^{-16}$. Linear regression for snowmobile data between bridging index and S_k produced an $R^2 = 0.50$ and $p < 2.2 \times 10^{-16}$.

The data displayed in Figure 6 start with calculated 2D stress values based on the well-known Bousinesq equations (Das, 1983) for an elastic homogeneous material. These calculations were then calibrated to match the median values at 40 cm, 60 cm, 80 cm and 100 cm into the snow cover for the measured stress under a snowmobile. A smoothing function was applied to the calibration factors to create a smooth transition between the medians. Each day the snow cover was classified into one of three simplified snow cover hardness profiles, similar to the finite element work of Schweizer (1993). The simplified hardness profile is shown on the left in the figures.

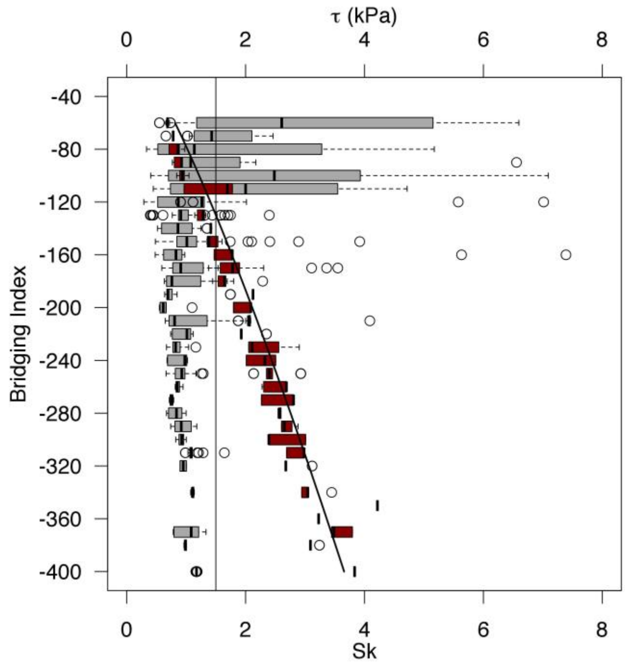

Figure 4: Box plots of skier shear stress (grey) and shear strength of surface hoar (red) by bridging index. The shear stress data are the converted measurements of normal stress added to the calculated shear stress due to the overlying slab. The diagonal line is the linear regression for bridging index and the calculated skier stability index. The vertical line corresponds to Sk = 1.5. Boxes, whiskers and open circles as in Figure 2.

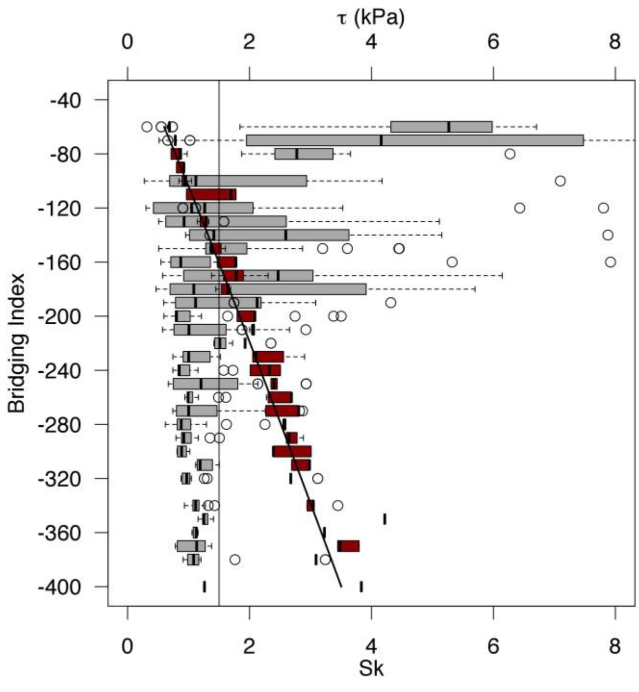

Figure 5: Box plots of snowmobile shear stress (grey) and shear strength of surface hoar (red) by bridging index. The shear stress data are the converted measurements of normal stress and calculated shear stress due to the overlying slab. The diagonal line is the linear regression for bridging index and the calculated skier stability index. The vertical line corresponds to Sk = 1.5. Boxes, whiskers and open circles as in Figure 2.

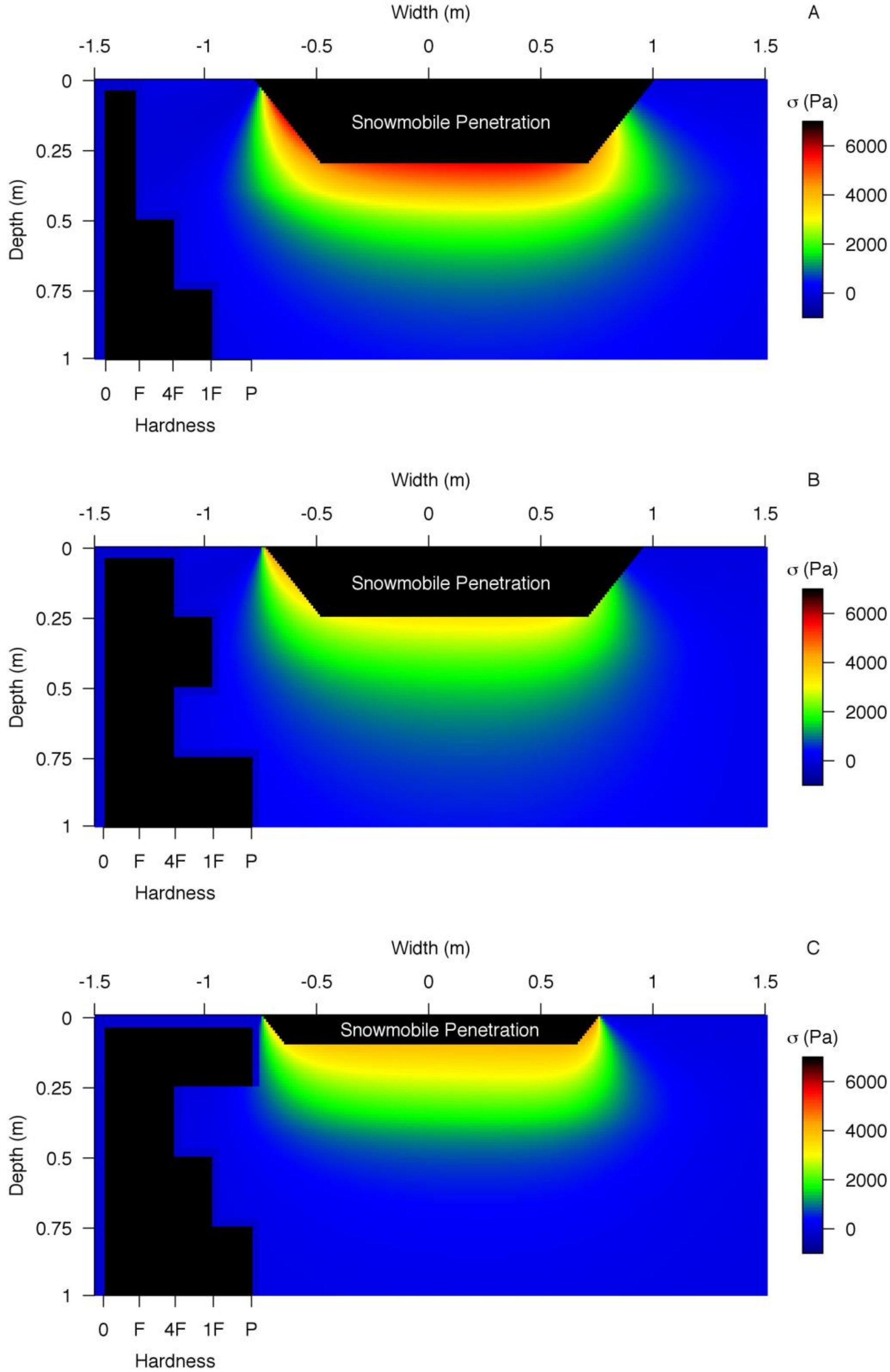

*Figure 6a, 6b and 6c: Calculated stress values for a static snowmobile applying stress to an elastic homogeneous snow cover. The calculated values were **calibrated** to match measured data for the passage of a snowmobile. Data shown were from days where the snow cover was classified according to the simplified hardness profile shown on the left in black. Plot A for "soft", B for "medium" and C for "supportive". The black near the surface of the plot represents the average snowmobile penetration into the snow cover.*

5. DISCUSSION

Figures 2 through 5 present data that could be used as a practical guide for estimating the impact from skiing and snowmobiling on a given snow cover. We must keep in mind that the data shown are measurements of dynamic movements through undisturbed snow! We drastically simplified the movements of the skier and snowmobile in order to make the experiments repeatable on subsequent days. Thumlert et al (2012) showed data from similar measurements that falling skiers added three to ten times more stress compared to a typical skier. Further, the variability in measured stress observed in Figures 2 through 5 show that caution should be used when interpreting these results.

The bridging index combined hardness of the snow cover and depth. Previous work confirmed the strong decrease of stress with increasing depth into the snow cover (Föhn 1987; Schweizer et al., 1993; Camponovo and Schweizer 1997; Jones et al. 2006; Habberman et al., 2008; Thumlert et al., 2013), so the strong decrease of stress with increased bridging is expected. The measurements also agree with the findings of Camponovo and Schweizer (1997) that harder layers create a bridging effect which spreads stress laterally and decreases the depth to which stress penetrated.

What is not shown in these plots are the stress concentrations that should occur at the interfaces between hard and soft layers. These concentrations have been shown in the finite element modelling work by Schweizer (1993), Jones et al. (2006) and Habermann et al. (2008). Skier triggered fractures in layers that produce avalanches generally occur at the interface between hard and soft layers (Schweizer and Jamieson, 2003). This is likely a combination of stress concentrating at these interfaces AND the weaker shear strength of the softer snow.

A recurring question is how much of what type of snow do you need to effectively bridge persistent weak layers? Reiweger and Schweizer (2010) published data that suggested weak layers of surface hoar are weaker in shear than in compression. Although it does not appear that science is settled on whether the initial failure in a weak layer is in shear (van Herwijnen et al., 2010), we used that assumption to relate bridging and slope stability. Jamieson (1995) showed that most dry slab avalanches occur when $S_{k38} < 1.5$. Investigating the regression lines in Figures 4 and 5 we see that $S_k = 1.5$ for a bridging index of 130 and 160 for skier and snowmobile data, respectively. However, the variability in S_k data means that a larger bridging index is required to ensure that $S_k > 1.5$. For all $S_k > 1.5$ bridging indices of 190 and 260 are required for skier and snowmobile data, respectively. Schweizer and Jamieson (2003) produced a median value for bridging of 110 for 89 slopes that were skied and did not slide.

What is a bridging index of 130? It can represent an infinite number of hardness profiles, but here are some examples:

- 50 cm fist, 40 cm 4F
- 20 cm P, 20 cm 4F
- 10 cm F, 20 cm 4F, 30 cm 1F

Figures 6a, b and c were created for two main purposes; to help visualize the bulb of stress created below localized loading and to observe the differences between varying snow cover structures. In his finite element calculations, Schweizer (1993) showed how the maximum additional shear stress $\Delta\tau$ by a skier decreased with increasing depth for three “simplified hardness profiles”. The data in Figure 6c agreed with Schweizer (1993) in that less stress below the supportive crust was observed compared to 6a and 6b.

It is important to understand that Figures 6a, b and c display calculations for stress that were calibrated to actual measured values. These measured values are assumed to be the peak stress added from the loading. Thus, the observed depth of the stress bulbs are showing measured values, but the widths were based on calculations for an elastic material. We would expect to see wider bulbs than observed for the supportive crust data in Figure 6c. Shapiro et al. (1997) wrote that the edges of the pressure bulb below vehicles moving in snow would maintain essentially vertical sidewalls that are aligned exactly with the lateral boundaries of the running gear (wheels). This suggests that the width of the stress bulb should not extend very far past the edges of the localized load.

6. CONCLUSION

Measurements of stress beneath localized dynamic loads (skiers and snowmobiles) were performed within undisturbed mountain snow cover. Measured stress levels were plotted against a *bridging index*. The bridging index combined hardness and thickness of a given layer in the snow cover. Measured stress decreased strongly with increased bridging which agrees with previous work (Föhn, 1987;

Schweizer et al., 1993; Schweizer and Camponovo, 1997; Jones et al., 2006; Habermann et al., 2008). The R^2 values for nonlinear regressions fit to measured stress against the bridging index were higher than those for measured stress against depth into the snow cover. Bridging indices of 130 (skier data) and 160 (snowmobile data) were found for calculated skier stability index of 1.5. These values should be used cautiously when relating to skier / snowmobile triggering.

The plots of calculated stress that were calibrated to stress measurements beneath a snowmobile show that softer snow allows more stress to penetrate. The plots helped to visualize the stress bulb beneath localized dynamic loads.

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